Frequency distribution tables

By size and month

Women shoe sales

	Germany, GER1												Germany, GER2												Mean	
US	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**GER1**	**GER2**
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00	0.00
4.5	0	0	0	0	1	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0.42	0.08
5	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0.17	0.17
5.5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	2	0	1	0.08	0.33
6	0	2	0	0	0	0	0	0	0	0	0	0	0	1	3	1	2	0	0	0	0	0	0	0	0.17	0.58
6.5	3	3	1	2	1	0	2	0	2	1	3	4	2	0	2	1	1	2	0	1	2	1	3	0	1.83	1.25
7	0	3	3	4	1	0	1	0	2	0	0	1	0	0	0	4	1	3	1	1	1	3	1	4	1.25	1.58
7.5	1	2	4	1	2	6	4	3	5	8	2	1	2	1	1	3	2	7	9	8	14	8	6	3	3.25	5.33
8	6	#	3	9	1	3	6	8	3	12	3	9	13	6	5	13	5	3	11	6	6	9	8	3	6.08	7.33
8.5	10	#	10	7	14	4	7	7	4	8	7	9	8	5	10	4	5	5	9	7	3	7	9	8	8.08	6.67
9	1	3	8	6	3	1	4	4	0	2	4	2	5	2	2	9	3	1	1	7	2	1	4	2	3.17	3.25
9.5	4	1	2	1	2	2	2	4	5	2	3	2	0	1	1	0	1	2	2	1	7	2	4	2	2.50	1.92
10	0	1	1	1	1	1	3	1	0	0	0	1	0	1	1	0	0	0	2	3	0	2	0	0	0.83	0.75
10.5	1	0	0	0	2	2	4	1	0	3	1	1	0	2	0	0	0	1	0	0	0	0	2	1	1.25	0.50
11.5	0	0	0	2	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	1	0	0	0	0	0.17	0.50
Total	**26**	**#**	**32**	**33**	**28**	**22**	**35**	**28**	**21**	**36**	**25**	**30**	**30**	**19**	**30**	**35**	**20**	**24**	**35**	**38**	**35**	**36**	**37**	**24**		

To find the confidence interval for the difference between two population means ($\mu_1 - \mu_2$) when the samples are independent, the variances are unknown, but we assume they are equal ($\sigma_1^2 = \sigma_2^2$), we use the **Pooled Variance t-interval**.

Since we assume the variances are equal, we "pool" (combine) the sample variances to get a better estimate of the shared population variance.

Sample variance		Pooled variance	Margin of error	95% CI	
GER1	GER2				
0.00	0.00	0.00	0.00	0.00	0.00
0.81	0.08	0.45	0.57	-0.23	0.90
0.33	0.33	0.33	0.49	-0.49	0.49
0.08	0.42	0.25	0.43	-0.68	0.18
0.33	0.99	0.66	0.69	-1.11	0.27
1.61	0.93	1.27	0.95	-0.37	1.54
2.02	2.27	2.14	1.24	-1.57	0.91
4.93	16.06	10.50	2.74	-4.83	0.66
12.27	12.24	12.25	2.96	-4.21	1.71
7.72	4.97	6.34	2.13	-0.72	3.55
5.06	6.57	5.81	2.04	-2.13	1.96
1.55	3.72	2.63	1.37	-0.79	1.96
0.70	1.11	0.91	0.81	-0.72	0.89
1.66	0.64	1.15	0.91	-0.16	1.66
0.33	2.09	1.21	0.93	-1.27	0.60

	GER1	GER2
n	12	12
$t_{22,0.025}$		2.07

The confidence interval is calculated as:

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Where:

- $\bar{x}_1$ and $\bar{x}_2$ are the sample means.
- n_1 and n_2 are the sample sizes.
- t^* is the critical value from the t-distribution table.
- s_p^2 is the **pooled variance**.

The pooled variance is a weighted average of the two sample variances (s_1^2 and s_2^2):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$df = n_1 + n_2 - 2$$